

Brian Kang

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Education

Robotics | Engineering Science - University of Toronto

Sep 2022 - Apr 2027

- **3.71 GPA** - *Dean's Honour List*

Experience

Trackside Engineering Placement - Red Bull Powertrains F1

July 2025 - Present

Product Validation & Data Science Engineer

- Development of a lumped-parameter **thermal model of the 2026 MGU-K**, deployed **on the race ECU** achieving **± 1 °C temperature prediction** live across all 4 cars to satisfy **FIA regulatory requirements**.
- Performed **competitor analysis** of energy deployment strategies, **reverse-engineering battery usage profiles** from onboard audio via **FFT and spectrogram analysis** to inform lap simulation and race strategy.
- Built a **genetic algorithm optimizer** to determine **laptime-optimal power unit allocation** across a full race season, updating dynamically as races occur to continuously re-optimize component usage.
- Authored **200+ lap dyno test specifications** for **power unit dyno running**, defining circuit selections, lap profiles and coolant configurations to generate datasets required for modelling.
- Developed a **Streamlit telemetry visualization platform**, deployed on the company **Kubernetes cluster**, enabling engineers across the factory to interactively generate custom plots from dyno data.

Power Unit Engineer

- Operated **Formula 1 power units on dynamometer test cells** to conduct durability testing and new engine pass-off programs, **monitoring real-time telemetry across engine, turbo, motor, and battery** systems to ensure parameters remain within performance and reliability envelopes.
- Diagnosed faults and anomalies during dyno running, **investigating issues across all subsystems** to trace root causes, coordinating with designers to determine solutions, and **developing diagnostic methodologies** for identifying and responding to recurring issues.
- Owned the **power unit operational limits document** defining safe operating boundaries across all systems, used as the reference by factory and trackside teams at every race weekend.

Projects

Machine Learning Soccer Analytics Tracker

July 2024 - Aug 2024

- Developed a computer vision model using **PyTorch, OpenCV, and YOLOv5** to perform object detection and classification of players and ball, achieving **95% detection accuracy**
- Implemented tracking and analysis algorithms to compute **team possession metrics and player speed** from detected objects.

Colour Sorting Robotic Arm

May 2023 - Aug 2023

- Built a 4-DOF robotic arm with closed-form inverse kinematics control for colour-based object sorting.
- Designed the full system including **mechanical design and fabrication** (3D Printing, Lasercutting), **electrical circuits and wiring**, and embedded **control software** for a microcontroller-based system.

Skills

Programming: Python, C/C++, Java, MATLAB, Simulink

ML & Robotics: PyTorch, TensorFlow, ROS

Embedded Systems: ESP32, RISC-V Assembly, FPGA, SystemVerilog, Quartus, ModelSim

CAD: Solidworks, Fusion360, AutoCAD, KiCad, SPICE

Hardware: 3D Printing, Laser Cutting, Oscilloscope, Soldering, Machining

F1: ATLAS, SystemMonitor